



EC MINOR PROJECT

ABSTRACT



TEAM MEMBERS:

SIDHARTH M

AMRUTHA M A

MARIYA MARGRET VICTOR

Intelligent Load Prioritization & Power Shedding System

PROBLEM STATEMENT

In residential and small commercial electrical systems, overload conditions often cause complete power shutdown due to circuit breaker tripping. Conventional systems lack intelligence to differentiate between critical and non-critical loads, resulting in unnecessary interruption of essential appliances.

PROPOSED SOLUTION

The proposed system develops an intelligent load management system that continuously monitors power consumption and automatically disconnects low-priority loads during overload conditions, while ensuring uninterrupted operation of high-priority loads.

TECHNICAL METHODOLOGY

- Current sensors are used to measure the current drawn by individual load sections.
- The microcontroller processes sensor data and calculates total power consumption.
- A priority-based algorithm is implemented to decide which loads should remain active.
- Relay modules are controlled to disconnect or reconnect loads based on priority and load conditions.
- System status can be displayed using an LCD.

COMPONENTS REQUIRED

- Arduino / ESP32 microcontroller
- Current sensors (ACS712)
- Relay modules
- Electrical loads (bulbs for demonstration)

- LCD display (optional)
- Power supply unit

EXPECTED OUTCOME

The system prevents complete power shutdown during overload conditions, improves power reliability, and demonstrates intelligent decision-making in embedded systems. It serves as a simplified model of smart load management used in modern power distribution systems.

Smart Power Factor Monitoring & Compensation System

PROBLEM STATEMENT

Low power factor in electrical systems results in inefficient power utilization, increased losses, and reduced system performance. In small residential, laboratory, and educational setups, power factor is rarely monitored due to the absence of simple and cost-effective solutions. Existing power factor correction systems are complex and unsuitable for small-scale or academic demonstrations.

PROPOSED SOLUTION

The proposed system designs a smart embedded power factor monitoring and compensation system that continuously measures voltage and current to estimate the power factor. When the power factor falls below a predefined threshold, the system automatically switches a capacitor using a relay to improve it, thereby demonstrating a simplified and cost-effective power factor correction model.

TECHNICAL METHODOLOGY

- Voltage and current sensors are used to acquire real-time electrical parameters.
- The microcontroller processes sensor signals to estimate the phase difference between voltage and current.
- Power factor is calculated using the phase angle relationship.
- When the power factor drops below a set value, a relay-controlled capacitor is switched ON to improve the power factor.
- The current power factor and system status are displayed on an LCD.

COMPONENTS REQUIRED

- Arduino / ESP32 microcontroller.
- Voltage sensor module.
- Current sensor (ACS712).
- Relay module.

- Capacitor (low-value, safe rating).
- LCD display.
- Electrical load (bulb for demonstration).
- Power supply unit.

EXPECTED OUTCOME

The system provides real-time power factor monitoring and automatic correction in a safe, low-voltage environment. It demonstrates efficient power utilization concepts, improves understanding of reactive power compensation, and serves as a compact educational model for embedded power systems.

Intelligent Access Control with Adaptive Timeout (Anti-Tailgating System)

PROBLEM STATEMENT

Conventional access control systems such as RFID or keypad-based doors do not address tailgating, where multiple individuals enter using a single authentication. This compromises security in restricted areas and laboratories.

PROPOSED SOLUTION

The proposed system designs an intelligent access control system that allows entry only after valid authentication and automatically enforces a time-limited access window. The system detects unauthorized attempts and restricts access, preventing tailgating and improving security.

TECHNICAL METHODOLOGY

- User authenticates using RFID or keypad.
- Door unlocks for a fixed time window.
- IR sensor checks entry count.
- If misuse is detected, system locks and alerts.

Authentication → Controller → Time Logic → Door Control → Alert.

COMPONENTS REQUIRED

- Arduino / ESP32.
- RFID reader or keypad.
- IR sensor.
- Servo motor (door model).
- Buzzer / LED.
- Power supply.

EXPECTED OUTCOME

The system improves access security by preventing unauthorized multiple entries, demonstrates embedded decision-making, and provides a low-cost solution suitable for restricted environments.